PIGNN-UQ: Physics-Informed Graph Neural Networks with Uncertainty Quantification for Registry-Compatible Blue Carbon Crediting

Nahla Nabil Skaik
Faculty of Computer Studies
Arab Open University
Kingdom of Bahrain
Email: nahla.nabil.52@gmail.com

***Abstract*—Mangrove carbon credits are discounted by a non-permanence buffer that does not depend on how uncertain the underlying carbon-stock estimate is. In this study, PIGNN-UQ combines a physics-informed graph neural network with gradient-boosted trees to map soil organic carbon over two southwest Florida mangrove regions (10,124 and 22,326 nodes of 150 m), and conformal prediction turns each estimate into a 90% interval, calibrated on held-out global-map labels, that feeds the uncertainty deduction of the Verra VM0033 methodology. The mean of the physics-informed network and XGBoost, selected by spatial cross-validation, reached a test R^2 of 0.28 and 0.60 (declared second test read), against 0.20 and 0.59 for XGBoost. The intervals covered 91.5% and 89.1% of test nodes, and their width tracked absolute error (Spearman 0.54 in the first region). The registry rule withheld a median 12.7% and 2.1% of credits per node, of the same order as the 11.0% median buffer of the seven issuing blue carbon projects; the label map's own uncalibrated intervals would withhold about 65%. The methodology's independent-error aggregation covered only 18% and 33% of realized project-level errors (upper 95% bounds 52% and 49%); empirical project-level calibration covered 91%. Against 87 soil cores at 25 sites, the ensemble had the lowest median error (10.1%, against 14.1% to 15.1% for two published maps and a regional mean), though within sampling uncertainty. These results motivate registry-scale tests of crediting rules that keep estimation uncertainty separate from non-permanence risk and aggregate it under spatially correlated errors.**

***Index Terms*—blue carbon, graph neural networks, conformal prediction, carbon credits, uncertainty quantification**

I. Introduction

Under Verra's VM0033 methodology, a mangrove project deposits part of its credits in a pooled buffer account against non-permanence risk (carbon later released through erosion, sea-level rise or disturbance), and a separate deduction applies when the 90% confidence interval of the estimate exceeds 20% of its value [1, p. 64]. The buffer share comes from a risk assessment and does not depend on how well the carbon stock itself was estimated. Published mangrove carbon maps either report no per-site interval or report intervals whose coverage and crediting effect have not been checked [2], [3], and the methodology combines plot uncertainties under an independence assumption that spatial models rarely satisfy.

This paper asks whether calibrated, per-site uncertainty from a spatial model can supply that deduction in a way a registry could audit. The proposed pipeline, PIGNN-UQ (Fig. 1), combines a physics-informed graph neural network (GNN) that encodes neighborhood structure and a tidal-inundation prior, tree ensembles as strong tabular baselines, and conformalized quantile regression for intervals calibrated on held-out data. This study departs from earlier work in four ways: (1) the interval of every node is converted into the VM0033 deduction, kept separate from the non-permanence buffer and compared with the buffers held by issuing projects; (2) the methodology's own aggregation rule is tested empirically and shown to understate project-level uncertainty when model errors are spatially correlated; (3) predictions and published maps are checked against 87 independent field soil cores; and (4) the analysis covers two regions, two image dates and leakage-free spatial splits, with model choices fixed on validation data under rules committed in advance.

The scope is deliberately applied: the paper does not propose a new network architecture. The contribution is a crediting pipeline whose uncertainty is calibrated, auditable and conservative, together with evidence on where that uncertainty is and is not informative. Table I positions this contribution against prior work.

II. Related Work

Physics-informed and graph learning. Physics-guided learning adds scientific structure to data-driven models [4]. Physics-informed networks penalize violations of governing equations [5], and graph formulations have carried such constraints into flood forecasting [6]. GraphSAGE [7] learns from node neighborhoods inductively, which suits spatial grids split into separate regions.

Mangrove carbon mapping. Most mangrove carbon studies regress field or map values on spectral indices with random forests or boosted trees [8]. Global soil carbon products are themselves machine-learning outputs [9], whereas regional maps such as the southwest Florida map of Radabaugh et al. are calibrated on local field cores. Geomorphic setting and tidal regime control mangrove soil carbon [10], [11], and soil

TABLE I
REQUIREMENTS OF MODEL-BASED CREDITING COVERED BY PRIOR WORK

Strand of work	Interval	Rule	Aggregation	Field	Market
Mangrove carbon maps	○	×	×	✓	×
Conformal prediction in EO	✓	×	×	×	×
Registry methodologies	×	✓	×	✓	×
Blue carbon market studies	×	×	×	×	✓
This study	✓	✓	✓	✓	✓

Interval: calibrated per-site interval (○: reported, not calibrated);
Rule: registry deduction; Aggregation: tested on realized errors;
Field, Market: checked against cores, registry records.

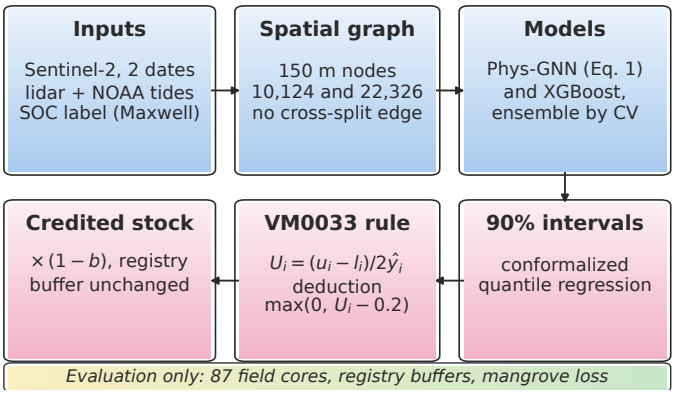


Fig. 1. PIGNN-UQ crediting pipeline. Field cores, registry records and mangrove loss are used only for evaluation.

carbon maps can disagree substantially at fine scales [12]. Random cross-validation overstates accuracy on such data, so spatial blocking is required [13], [14].

Uncertainty quantification. Split conformal prediction gives finite-sample coverage under exchangeability [15], and conformalized quantile regression adapts the interval width to each input [16] and has been applied to earth observation [17]. Spatial dependence weakens exchangeability, which motivates localized (Mondrian) calibration [18], [19], whose validity needs dense calibration data in each region [20]. Interval quality is scored with the Winkler interval score [21], [22].

Crediting and markets. Blue carbon credits traded at a weighted average of US$26.96 per credit in 2022 and 2023 [23], and registry records report issuances, retirements and buffer deposits per project [24].

Research gap. Table I compares these strands against five requirements of model-based crediting. No study, to our knowledge, (i) converts calibrated spatial intervals into a registry uncertainty deduction, (ii) tests the registry's aggregation rule against realized errors, and (iii) checks the result against field cores and market records at the same time.

III. DATA AND METHODS

A. Regions, data and graph

Two southwest Florida mangrove regions were used: Ten Thousand Islands / Rookery Bay (R1; $-81.90°$ to $-81.40°$, $25.90°$ to $26.30°$) and the Everglades National Park coast (R2; $-81.20°$ to $-80.90°$, $25.30°$ to $25.60°$). The label is the 30 m soil organic carbon stock (0 to 100 cm, t C/ha) of the 2020 global mangrove product of Maxwell et al., which also provides per-pixel 95% prediction limits. Native cells are aggregated into 5×5 blocks (about 150 m) kept where the Global Mangrove Watch 2020 fraction exceeds 0.5 [25], giving 10,124 nodes (64,366 directed 8-neighbor edges) in R1 and 22,326 nodes (143,034 edges) in R2. Node features are ten Sentinel-2 L2A bands and derived indices [26] for 6 February 2026 and, with cloud masking, for 15 August 2026, together with elevation and distance to the tide gauge. While revising this work, the Sentinel-2 clip of the earlier version was found shifted 9.18 km north (mean normalized difference vegetation index (NDVI) of mangrove nodes 0.003); the shift was reproduced on correctly referenced imagery (correlation 0.91 to 0.97), and all features were rebuilt (mean NDVI 0.46).

Splits are spatial: fine cells of about 2.5 km are assigned whole to training, validation or test (R1: 6,465/1,583/2,076 nodes; R2: 14,499/3,351/4,476), and every model is trained and evaluated on subgraphs induced by one split, so no edge crosses a split boundary.

B. Models and physics prior

Baselines are linear regression, a location-only random forest, random forest with and without neighbor-mean (spatial lag) features, and XGBoost [27], [28]. The graph model is a two-layer GraphSAGE network (hidden size 16, dropout 0.3, Adam, learning rate 0.01, early stopping on held-out groups). The physics-informed variant adds a steady-state prior,

$$\mathcal{L} = \mathrm{MSE}(\hat{y}, y) + \lambda\, \mathrm{MSE}(\hat{y}_z, P_z) + w\, \overline{\mathrm{ReLU}(-\hat{y})^2}, \quad (1)$$

where $\hat{y}_z$ and P_z are the standardized prediction and prior, $\lambda = 0.1$ (chosen by CV from 0.1, 0.3 and 1) and $w = 0.1$ weight the prior and non-negativity terms, and P is the mean of min-max scaled tidal inundation frequency (the share of 2025 hourly National Oceanic and Atmospheric Administration (NOAA) water levels above each node's U.S. Geological Survey (USGS) 3DEP lidar elevation, both in NAVD88) and red-edge canopy index (NDRE), reflecting the role of hydroperiod in sediment delivery and carbon preservation. The prior acts as a soft domain constraint rather than a governing equation, and it replaced the NDVI and gauge-distance proxy of the earlier version. The final candidate set added the unweighted mean of the physics-informed GNN and XGBoost (the ensemble), a stacked variant in which the GNN corrects out-of-fold XGBoost predictions, and a GNN trained on both image dates. Table II lists every candidate with its accuracy.

C. Uncertainty and crediting

Quantile models (GNN, physics-informed GNN, XGBoost), normalized conformal random forest and a quantile ensemble were calibrated by split conformal prediction on the validation split, either globally or per spatial block (Mondrian, at least 30 calibration nodes per block). The trivial interval is the training mean $\pm$ the conformal quantile of $|y - \bar{y}|$ on the same calibration nodes, and deductions are clipped to [0, 1]. For node i with estimate $\hat{y}_i$ and 90% interval $[l_i, u_i]$, the relative uncertainty $U_i = (u_i - l_i)/(2\hat{y}_i)$ enters the VM0033 rule,

$$c_i = \hat{y}_i \left(1 - \max(0,\, U_i - 0.20)\right)(1 - b), \quad (2)$$

TABLE II
MODEL ACCURACY (R^2) AND FIELD-SITE ERROR

Model	R1 CV	R1 test	R2 test	Field (%)
Constant (train mean)	n/a	-0.11	-0.01	15.0
Linear	0.22	-0.24	0.41	n/a
Forest on lon/lat only	0.42	-0.79	0.57	n/a
Random forest	0.38	0.03	0.58	n/a
XGBoost	0.41	0.20	0.59	16.5
GNN	0.39	0.07	0.58	n/a
Physics-informed GNN	0.41	0.17	0.56	11.6
Ensemble (GNN+XGBoost)	**0.47**	**0.28**†	**0.60**†	**10.1**

Test: mean of 3 seeds. †Second, declared test read. Field: median absolute error at 25 core sites; published maps 14.1 and 15.1.

where b is the registry's non-permanence buffer, applied unchanged. Interval quality and deductions are summarized in Table III and mapped per node in Fig. 3.

D. Evaluation protocol

Feature sets, model settings, the physics prior and the interval method were chosen by spatial five-fold cross-validation (CV) on training and validation nodes, using selection rules written and committed before each experiment. The test split was read once for all pre-specified models. After that read, one further model (the ensemble) was pre-specified on cross-validation alone and evaluated in a second, declared test read, and its test numbers are marked in Table II. Analyses that followed the first test read (generalization, two-date training, project aggregation, field cores, market data) used validation data or external data only and are reported as exploratory. Nine logged deviations and all rules are archived with the code.

IV. EXPERIMENTS AND RESULTS

This section reports measurements only; their interpretation is deferred to Section V.

A. Accuracy

Table II and Fig. 2(a) report accuracy. In R1 the ensemble reached test $R^2 = 0.277 \pm 0.035$ (mean $\pm$ standard deviation (SD) over three seeds; mean absolute error (MAE) 58.1 t C/ha), against 0.203 for XGBoost and 0.170 for the physics-informed GNN, whose seed SD (0.054) was a quarter of the plain GNN's (0.236). A constant training mean scored -0.11, and the location-only model -0.79 despite a cross-validated R^2 of 0.42, so location alone did not transfer to unseen blocks. In R2 all non-linear models scored between 0.56 and 0.60. On interior test nodes at least two hops from any other split, the ensemble kept $R^2 = 0.19$ (R1) and 0.58 (R2). With all other settings fixed, correcting the image georeferencing alone raised the GNN's cross-validated R^2 from 0.17 to 0.26, and adding the prior raised it from 0.39 to 0.41, within one standard error (0.03). The physics prior's correlation with the predictions it constrains was 0.14 on R1 test nodes, against 0.76 for the earlier proxy.

B. Generalization and stability

A single ensemble trained on both regions reached cross-validated $R^2 = 0.37$ (R1) and 0.53 (R2), against 0.47 and 0.55 for region-specific models, whereas models applied to a region they had not seen scored below zero (Fig. 2(b)). In that zero-shot setting, conformal intervals calibrated on the source region covered 76% to 89% of target validation nodes, and recalibrating them on half of the target's validation nodes restored 88% to 97%. Training the physics-informed GNN on both image dates raised the rank agreement of its predictions across dates from 0.44 to 0.88.

TABLE III
INTERVALS (90%, TEST) AND CREDITING

	R1	R2
Coverage	91.5%	89.1%
Mean width / trivial (t C/ha)	273 / 274	191 / 282
Winkler score / trivial	306 / 372	247 / 369
ρ(width, abs. error)	0.54	0.15
Median VM0033 deduction per node	12.7%	2.1%
Project coverage, independent errors*	18%	33%
Project cov., bound / empirical*	91 / 91%	100 / 91%

*Realized errors of 2.5 km pseudo-projects, validation data.

C. Intervals and crediting

Table III and Fig. 2(c) summarize the selected intervals (conformalized quantile GNN, global calibration). In R1 the intervals matched the trivial interval's width but scored 18% better on the Winkler score, and their width ranked absolute errors ($\rho = 0.54$, 95% confidence interval 0.51 to 0.58). In R2 they were 32% narrower than the trivial interval (191 vs. 282 t C/ha). Mondrian calibration covered 87.9% in R1, where only six validation blocks were available. Applied per node, the VM0033 rule withheld a median 12.7% (R1) and 2.1% (R2) of credits. Across all five interval methods, credited stock stayed within 0.84 to 0.90 (R1) and 0.95 to 0.98 (R2) of the buffer-only amount. As an illustrative standing-stock exposure, not credit issuance, the deducted share over the 2,076 R1 test nodes (about 4,700 ha) equals 0.71 million t CO_2e, or US$19.2 million at the 2022 and 2023 average credit price (Section II). This is an upper bound, because credits are issued on stock changes rather than on standing stock. Interval width did not track mangrove loss between 2010 and 2020 (Spearman -0.06 in both regions). Among the seven blue carbon projects with issued credits (6.62 million), the median buffer was 11.0% and 73.5% of credits had been retired (registry records, Section II).

When node uncertainties were aggregated to 2.5 km pseudo-projects as VM0033 prescribes (root-sum-of-squares, independent errors), the stated project uncertainty covered the realized aggregate error in 2 of 11 projects in R1 (18%, exact 95% interval 2–52%) and 15 of 45 in R2 (33%, 20–49%) and implied no deduction. A fully correlated bound covered 91% and 100% (median deductions 14.6% and 3.1%), and an empirical project-level conformal calibration on held-out pseudo-projects covered 91% in both regions.

D. Field soil cores

The 87 mangrove cores of the Coastal Carbon Library [29] were converted to stocks over the top 100 cm, using a quadratic organic-matter conversion of the form in Craft et al. [30] fitted on 679 regional samples ($R^2 = 0.91$), and matched to nodes within 300 m. At the 25 sites, the median absolute

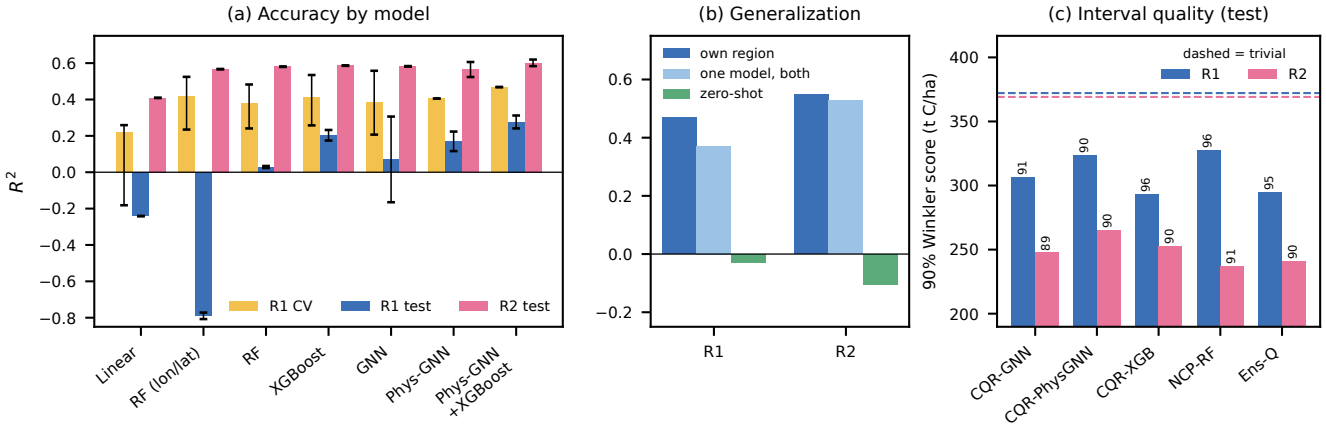


Fig. 2. (a) R^2 by model (R1 CV with fold range; test mean $\pm$ SD over 3 seeds). (b) Ensemble generalization. (c) 90% Winkler score on test; dashed: trivial interval; labels: coverage (%). CQR: conformalized quantile regression; NCP: normalized conformal; Ens-Q: quantile ensemble.

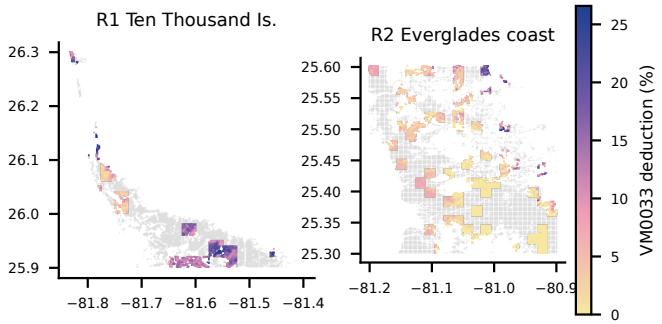


Fig. 3. Per-node VM0033 deduction (Eq. 2) on test nodes; gray: training and validation nodes.

error was 10.1% for the ensemble, 11.6% for the physics-informed GNN, 16.5% for XGBoost, 14.1% and 15.1% for the Maxwell and Radabaugh maps, and 15.0% for a constant. Paired bootstrap 95% intervals of these differences included zero (ensemble minus Maxwell map -4.0 points, -8.1 to 8.2), and the Maxwell map ranked sites best (Spearman 0.63; ensemble 0.29). The two published maps were uncorrelated at node level (-0.10 in R1). For the 13 cores sampled to at least 50 cm, field-calibrated intervals covered 85% to 92% under leave-one-study-out and implied VM0033 deductions of 22% to 29%. The label product's own 95% prediction limits contained all 13 cores, but their median half-width equaled 101% of the estimate. Rescaled to 90% under a normal approximation, they would imply a VM0033 deduction of about 65% in both regions.

V. Discussion

The physics-informed GNN and tree ensembles are complementary. Alone, the physics-informed GNN matched XGBoost in cross-validation (0.41 each), but their mean outperformed every other final candidate in cross-validation (0.47 and 0.55, against at most 0.42 and 0.53) and every model on test (Table II). The GNN adds neighborhood structure and the hydroperiod prior that the trees do not see, and it was the most stable graph model across seeds and, trained on both dates, across seasons. The physics prior now acts as a weak constraint rather than a target the model copies.

Estimation uncertainty is of the same order as the buffer the market already holds. The per-node deduction in R1 (12.7%) is similar in size to the realized median buffer of issuing blue carbon projects (11.0%), yet today it is not applied; the similarity is context, not validation, because the two cover different risks. Calibration is what makes the deduction usable, since the label map's uncalibrated limits would withhold about five times as much in R1 (Section IV-D). Fig. 3 shows that the deduction concentrates in specific areas, so it is a location-dependent quantity that a flat buffer cannot represent. Because width does not follow observed mangrove loss, the two discounts measure different risks and belong in separate factors, as in Eq. 2. **Business implication:** if validated on registry project data, a registry could adopt the deduction without changing its buffer policy, and an issuer can see in advance which areas carry the largest deduction and where additional field sampling would raise the credited volume most.

Aggregation under independent errors is the weak point of the registry rule. Spatial model errors are correlated, so root-sum-of-squares aggregation shrinks project uncertainty and grants no deduction even where realized errors exceed the allowance. Both a fully correlated bound and an empirical project-level calibration restored nominal coverage. Model-based crediting therefore needs an aggregation rule that accounts for spatial dependence.

What the intervals can and cannot do. Interval width is bounded by accuracy: with test $R^2 = 0.07$ for the GNN underlying the intervals in R1, at most about 4% narrowing over the trivial interval is attainable, and in R2 ($R^2 = 0.58$) the 32% achieved is close to the 35% bound. The value in R1 lies in adaptivity and Winkler score, not sharpness. Local calibration is required when a model is moved to a new region. Field-calibrated deductions exceed label-calibrated ones, a reminder that the label product carries its own error, in line with the fine-scale disagreement between soil carbon maps

reported for United States tidal wetlands.

VI. Limitations

Labels are a modeled product: the training label is a global map that disagrees with a field-calibrated regional map at node level. **Accuracy in R1 is moderate** ($R^2 = 0.28$). **The ensemble was evaluated in a second test read,** declared and limited to one pre-specified model. **Single tide gauges** represent each region, and R2's is about 70 km away. **Field cores are sparse** (13 deep enough for 100 cm stocks) and may overlap the global map's training data. **Market outcomes** were not modeled, since prices and buffers come from published records.

VII. Conclusion

Per-node uncertainty from a physics-informed GNN and XGBoost ensemble, calibrated on held-out map labels, can supply the VM0033 uncertainty deduction. The deduction (median 12.7% in R1) is of the same order as the 11.0% buffers that issuing projects hold, with interval coverage of 89% to 92% in two regions. The methodology's independent-error aggregation covered only 18% to 33% of realized project errors, and against 87 field cores the ensemble had the lowest site-level median error (10.1%), although the margin over published maps was within sampling uncertainty. Future work will test a correlation-aware aggregation with registry project data, add regions with denser field cores, and replace the steady-state prior with repeated elevation and accretion measurements.

Code Availability

Code, data pipeline, selection rules and results are available at https://github.com/Nahla-Nabil/PIGNN-UQ.

References

[1] Verra, "VM0033: Methodology for tidal wetland and seagrass restoration, v2.1," https://verra.org/methodologies/vm0033-methodology-for-tidal-wetland-and-seagrass-restoration-v2-1/, 2023.

[2] T. L. Maxwell *et al.*, "Global mangrove soil organic carbon stocks dataset at 30 m resolution for the year 2020 based on spatiotemporal predictive machine learning," *Data in Brief*, vol. 50, p. 109621, 2023, doi: 10.1016/j.dib.2023.109621.

[3] K. R. Radabaugh *et al.*, "A spatial model comparing above- and belowground blue carbon stocks in southwest Florida mangroves and salt marshes," *Estuaries and Coasts*, vol. 46, pp. 1536–1556, 2023, doi: 10.1007/s12237-023-01217-7.

[4] J. Willard, X. Jia, S. Xu, M. Steinbach, and V. Kumar, "Integrating scientific knowledge with machine learning for engineering and environmental systems," *ACM Computing Surveys*, vol. 55, no. 4, pp. 1–37, 2023, doi: 10.1145/3514228.

[5] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019, doi: 10.1016/j.jcp.2018.10.045.

[6] M. Taghizadeh, Z. Zandsalimi, M. A. Nabian, M. Shafiee-Jood, and N. Alemazkoor, "Interpretable physics-informed graph neural networks for flood forecasting," *Computer-Aided Civil and Infrastructure Engineering*, vol. 40, pp. 2629–2649, 2025, doi: 10.1111/mice.13484.

[7] W. L. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, arXiv:1706.02216.

[8] A. Dutta Roy *et al.*, "Remote sensing-based mangrove blue carbon assessment in the Asia-Pacific: A systematic review," *Science of The Total Environment*, vol. 938, p. 173270, 2024, doi: 10.1016/j.scitotenv.2024.173270.

[9] J. Sanderman *et al.*, "A global map of mangrove forest soil carbon at 30 m spatial resolution," *Environmental Research Letters*, vol. 13, p. 055002, 2018, doi: 10.1088/1748-9326/aabe1c.

[10] A. S. Rovai *et al.*, "Global controls on carbon storage in mangrove soils," *Nature Climate Change*, vol. 8, pp. 534–538, 2018, doi: 10.1038/s41558-018-0162-5.

[11] K. W. Krauss *et al.*, "How mangrove forests adjust to rising sea level," *New Phytologist*, vol. 202, pp. 19–34, 2014, doi: 10.1111/nph.12605.

[12] J. R. Holmquist *et al.*, "Accuracy and precision of tidal wetland soil carbon mapping in the conterminous United States," *Scientific Reports*, vol. 8, p. 9478, 2018, doi: 10.1038/s41598-018-26948-7.

[13] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, pp. 913–929, 2017, doi: 10.1111/ecog.02881.

[14] P. Ploton *et al.*, "Spatial validation reveals poor predictive performance of large-scale ecological mapping models," *Nature Communications*, vol. 11, p. 4540, 2020, doi: 10.1038/s41467-020-18321-y.

[15] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *Journal of the American Statistical Association*, vol. 113, pp. 1094–1111, 2018, doi: 10.1080/01621459.2017.1307116.

[16] Y. Romano, E. Patterson, and E. J. Candès, "Conformalized quantile regression," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, arXiv:1905.03222.

[17] G. Singh, G. Moncrieff, Z. Venter, K. Cawse-Nicholson, J. Slingsby, and T. B. Robinson, "Uncertainty quantification for probabilistic machine learning in earth observation using conformal prediction," *Scientific Reports*, vol. 14, p. 16166, 2024, doi: 10.1038/s41598-024-65954-w.

[18] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005, doi: 10.1007/b106715.

[19] X. Lou, P. Luo, and L. Meng, "GeoConformal prediction: A model-agnostic framework for measuring the uncertainty of spatial prediction," *Annals of the American Association of Geographers*, vol. 115, no. 8, pp. 1971–1998, 2025, doi: 10.1080/24694452.2025.2516091.

[20] H. Mao, R. Martin, and B. J. Reich, "Valid model-free spatial prediction," *Journal of the American Statistical Association*, vol. 119, no. 546, pp. 904–914, 2024, doi: 10.1080/01621459.2022.2147531.

[21] R. L. Winkler, "A decision-theoretic approach to interval estimation," *Journal of the American Statistical Association*, vol. 67, pp. 187–191, 1972, doi: 10.1080/01621459.1972.10481224.

[22] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, pp. 359–378, 2007, doi: 10.1198/016214506000001437.

[23] S. Farahmand, N. Hilmi, and C. M. Duarte, "The rise and flows of blue carbon credits advance global climate and biodiversity goals," *npj Ocean Sustainability*, vol. 4, p. 39, 2025, doi: 10.1038/s44183-025-00141-6.

[24] P. Quartson *et al.*, "Voluntary Registry Offsets Database v2026-08," https://gspp.berkeley.edu/berkeley-carbon-trading-project/offsets-database, 2026, Berkeley Carbon Trading Project, University of California, Berkeley.

[25] P. Bunting *et al.*, "Global mangrove extent change 1996–2020: Global Mangrove Watch version 3.0," *Remote Sensing*, vol. 14, p. 3657, 2022, doi: 10.3390/rs14153657.

[26] M. Drusch *et al.*, "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sensing of Environment*, vol. 120, pp. 25–36, 2012, doi: 10.1016/j.rse.2011.11.026.

[27] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[28] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[29] Coastal Carbon Network, "Database: Coastal Carbon Library (version 1.7.0)," Smithsonian Environmental Research Center, dataset, 2025, doi: 10.25573/serc.21565671, accessed 2026-10-09.

[30] C. B. Craft, E. D. Seneca, and S. W. Broome, "Loss on ignition and Kjeldahl digestion for estimating organic carbon and total nitrogen in estuarine marsh soils: Calibration with dry combustion," *Estuaries*, vol. 14, pp. 175–179, 1991, doi: 10.2307/1351691.